

The Impact of AI on jobs by 2030

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[Source: \(Kaggle\)](#)

[Github](#)

Project Overview

- I analyzed how AI will affect jobs over the next 4 years
- This was done with a dataset containing:
 - Job titles
 - Skills
 - Education levels
 - AI exposure
 - Automation probability
- Goal: Identify which jobs are most vulnerable and why

Key Research Questions

- Which professions face the highest automation risk
- How do skills influence job safety
- Does education protect against automation
- How does AI exposure relate to automation probability
- What patterns emerge across salary, experience, and risk categories

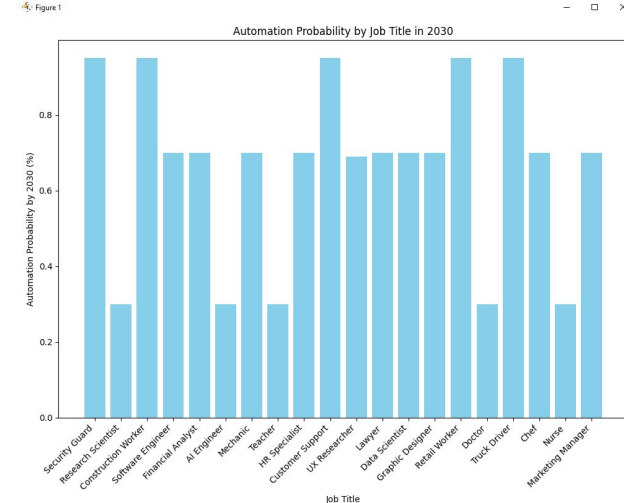
Dataset Description

- This dataset has data on 3000 jobs being tracked across 20 job titles.
- Variables include:
 - Average Salary
 - Years of Experience
 - Education Level
 - AI Exposure Index
 - Tech Growth Factor
 - 10 Skill Indicators
 - Automation Probability (2030)
 - Risk Category
- There are no missing values; and duplicates have been removed

Automation Probability by Job Title

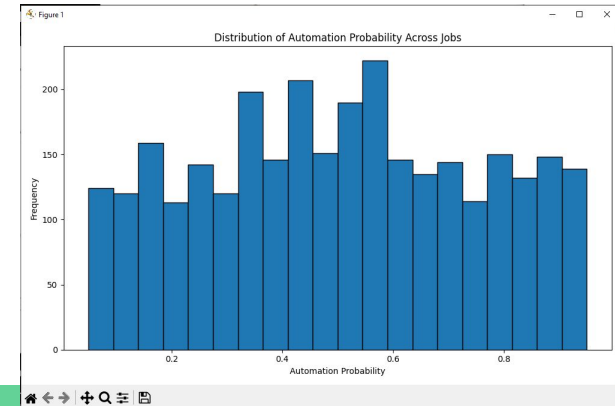
Key insights:

- Jobs like Customer Support, Retail Worker, Truck Driver, and Security Guard tend to have higher automation probabilities because they involve structured, predictable tasks.
- Jobs like Doctor, Nurse, Research Scientist, and AI Engineer show lower automation risk due to complex decision-making, human interaction, or specialized knowledge.
- The spread of probabilities suggests that AI will not impact all jobs equally — some sectors face far more disruption than others.



Automation Probability Distribution

- The histogram reveals how automation risk is spread across the entire job market.
- Wide spread of automation probabilities
- Indicates diverse vulnerability across job types
- The distribution appears fairly wide, meaning automation risk varies significantly between jobs.
- Peaks around mid-range values (e.g., 0.4–0.6) suggest many jobs face moderate automation risk.
- The presence of both low-risk and high-risk clusters indicates a polarized job market — some roles are very safe, others very vulnerable.
- This supports the idea that AI will reshape certain industries more dramatically than others.

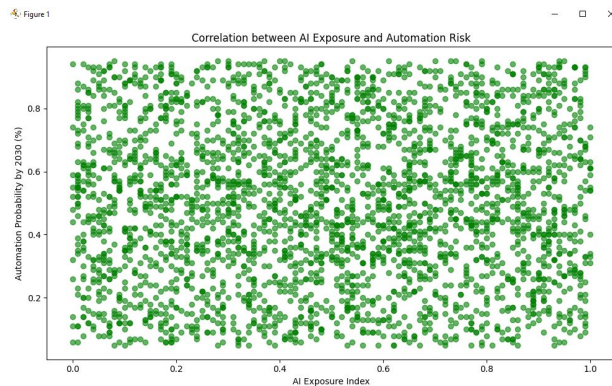


AI Exposure vs Automation Probability

- Some jobs with high AI exposure (e.g., AI Engineer, Data Scientist) actually have low automation risk — because they create or manage AI systems rather than being replaced by them.
- Some low-exposure jobs (e.g., Truck Driver, Retail Worker) still show high automation risk due to robotics and autonomous systems.
- The computed correlation value gives a more precise measure of the relationship. If the correlation is weak or moderate, that matches what the scatterplot shows visually.
- (AI Exposure vs Automation Probability) Correlation: 0.014319822852625513

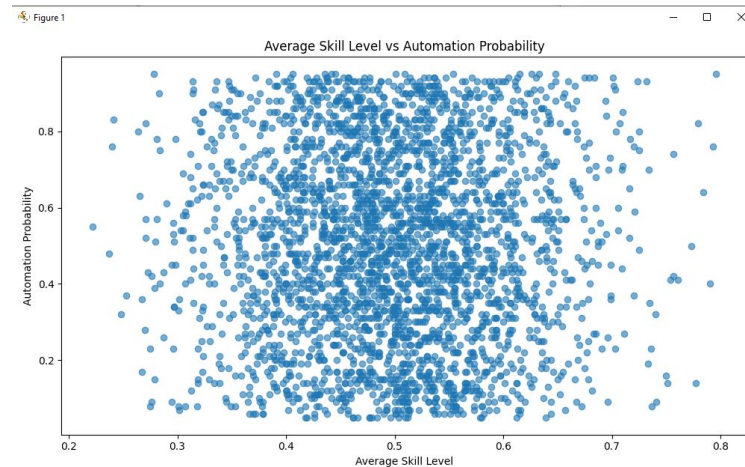
Correlation Formula
used:

$$r_{xy} = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i=1}^n (x_i - \bar{x})^2} \sqrt{\sum_{i=1}^n (y_i - \bar{y})^2}}$$



Skill Level vs Automation Probability

- Average skill level computed
- Negative relationship:
 - Higher skill means lower automation risk
- Skill-intensive jobs are harder to automate
- The points are scattered with no clear linear trend.
- Skill level alone does not strongly predict automation probability.
- There is a slight downward tendency (higher skill → lower risk), but it's weak.
- Why it looks messy:
 - Skill levels in the dataset are fairly close together (0.2–0.8).
 - Automation probabilities span the full range (0–1).
 - This mismatch naturally produces a cloud of points.



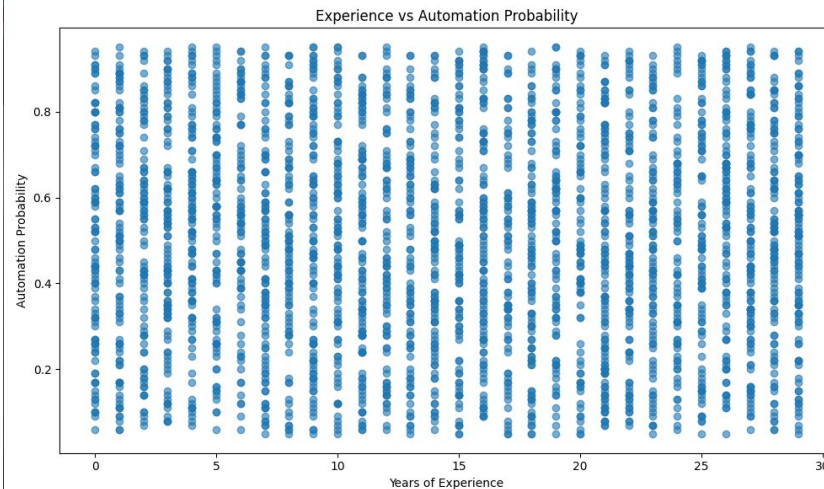
Experience vs Automation Probability

- There is no visible correlation between years of experience and automation probability.
- Jobs requiring 0 years of experience can be low-risk or high-risk.
- Jobs requiring 20+ years of experience can also be low-risk or high-risk.

Why it looks random:

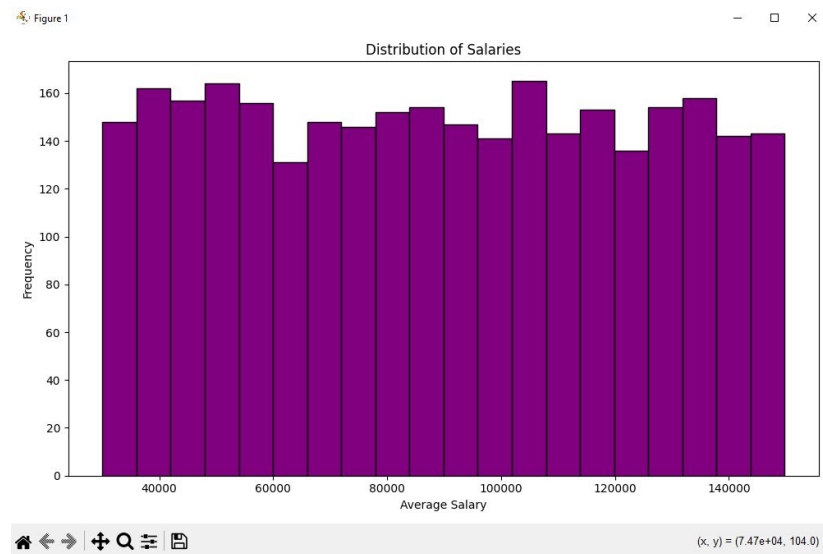
- Experience is not used in the dataset to determine automation risk.
- Automation probability is tied more to job type than to seniority.

Figure 1



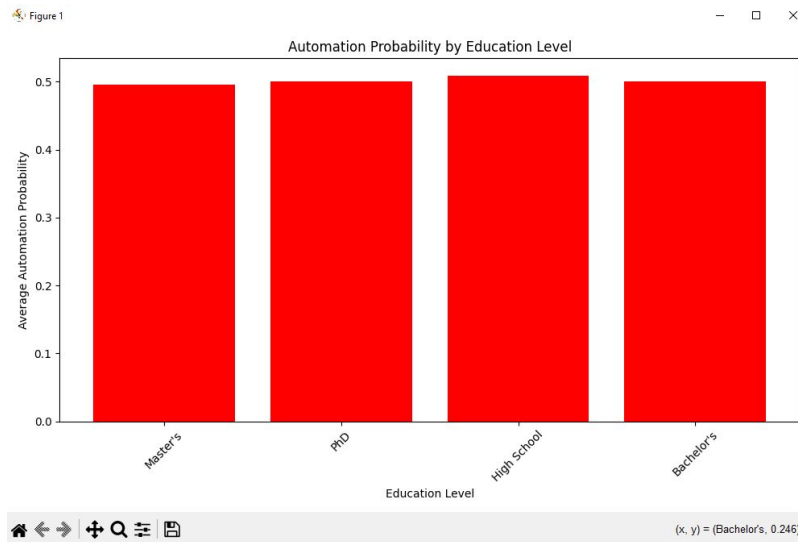
Salary Distribution

- Salaries in the dataset are fairly uniform across the \$40k–\$140k range.
- There is no strong concentration in any particular salary band.
- Because salaries do not vary dramatically, salary is not a strong predictor of automation risk.
- Why this matters:
 - A uniform salary distribution means the automation-risk analysis is not distorted by salary differences.
 - This helps isolate the effect of job characteristics, rather than pay, on automation risk.



Automation Probability by Education Level

- All education levels have **nearly identical** average automation probabilities (~ 0.50).
- High School jobs are only *slightly* higher, but the difference is negligible.
- Bachelor's, Master's, and PhD roles show **no meaningful separation**.
- The wide spread within each group indicates that **job type**, not education, determines automation risk.
- Education level is **not a predictive factor** in this dataset.



Top 10 Jobs Most at Risk (Averaged):

Job_Title	Automation_Probability_2030 (Average)
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Retail Worker	0.834710
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Security Guard	0.833052
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Construction Worker	0.825962
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Customer Support	0.825263
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Truck Driver	0.818627
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Graphic Designer	0.529000
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Software Engineer	0.508686
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Data Scientist	0.496287
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Chef	0.495674
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Marketing Manager	0.494776
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Top 10 Jobs Least at Risk (Averaged):

Job_Title	Automation_Probability_2030 (Average)
UX Researcher	0.492395
HR Specialist	0.490764
Lawyer	0.486439
Mechanic	0.485294
Financial Analyst	0.485099
Doctor	0.176742
Teacher	0.175783
Nurse	0.175634
Research Scientist	0.174161
AI Engineer	0.172214

Sample Mean and Population Mean

Population Mean

$$\mu = \frac{\sum_{i=1}^N x_i}{N}$$

Sample Mean

$$\bar{X} = \frac{\sum_{i=1}^n x_i}{n}$$

```
sample = df.sample(100)

sample_mean = mean(sample['Automation_Probability_2030'].to_list())
population_mean = mean(df['Automation_Probability_2030'].to_list())

print("\nSample Mean:", sample_mean)
print("Population Mean:", population_mean)
```

- Sample Mean: 0.484999999999999965
- Population Mean: 0.5015033333333333
- The averages between the entire population and a sample of 100 records is very close

Conclusion

- Automation risk varies widely across job types
- Routine jobs face the highest risk
- Human-centric and specialized roles are safest
- Salary and education level do *not* strongly predict automation risk
- Job characteristics (task structure, human interaction, creativity) matter most
- AI exposure alone does not determine vulnerability